MODULE 21

Advanced Intelligence Techniques

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

By the end of this module, you will chase a criminal through the network like a professional intelligence officer.

- How the network marks a criminal - **IP addresses, DNS records and traffic patterns**
- What **packet, proxy and interception** really mean, and how to use them in a case
- The intelligence vocabulary - **CVE, APT, VAPT and OWASP**
- Capturing your own traffic with **Wireshark** and catching tampering with **hashes**
- Why **encrypted evidence** must never be broken by you - the correct procedure

The AIIMS Attack: A Wake-Up Call

23 November 2022. Doctors at Delhi's AIIMS arrived to find **nothing opening** on their screens - appointment, billing, lab reports, all down.

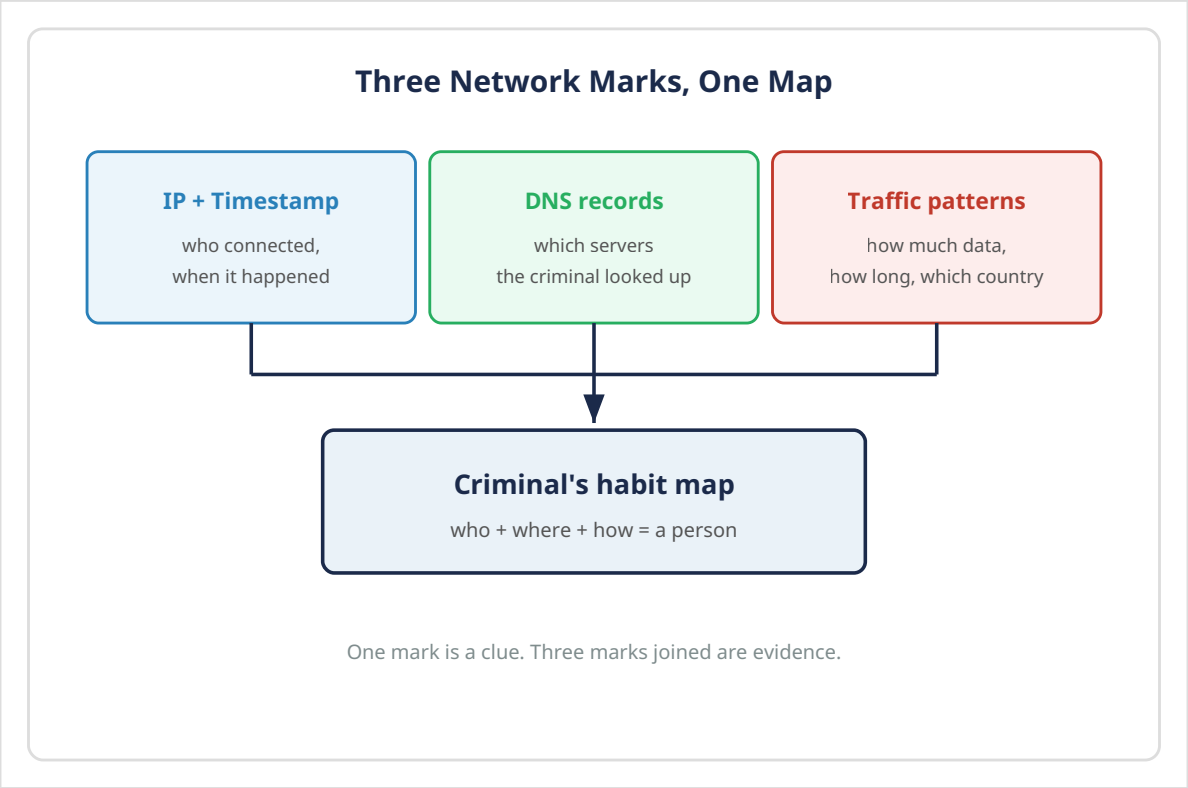
- The hospital's servers had suffered a **ransomware attack** - data encrypted and held hostage
- Investigators traced an attacker IP to **Hong Kong** - but experts warned criminals route through VPNs to hide
- CERT-In found **basic security gaps** - like missing firewall policies
- Lesson: the criminal no longer arrives with a knife. **He comes through the network** - so you must learn its language

Chasing the Criminal Through the Network

Network-based crime analysis means **tracking the criminal through his digital routes**. Every online crime leaves marks on the network.

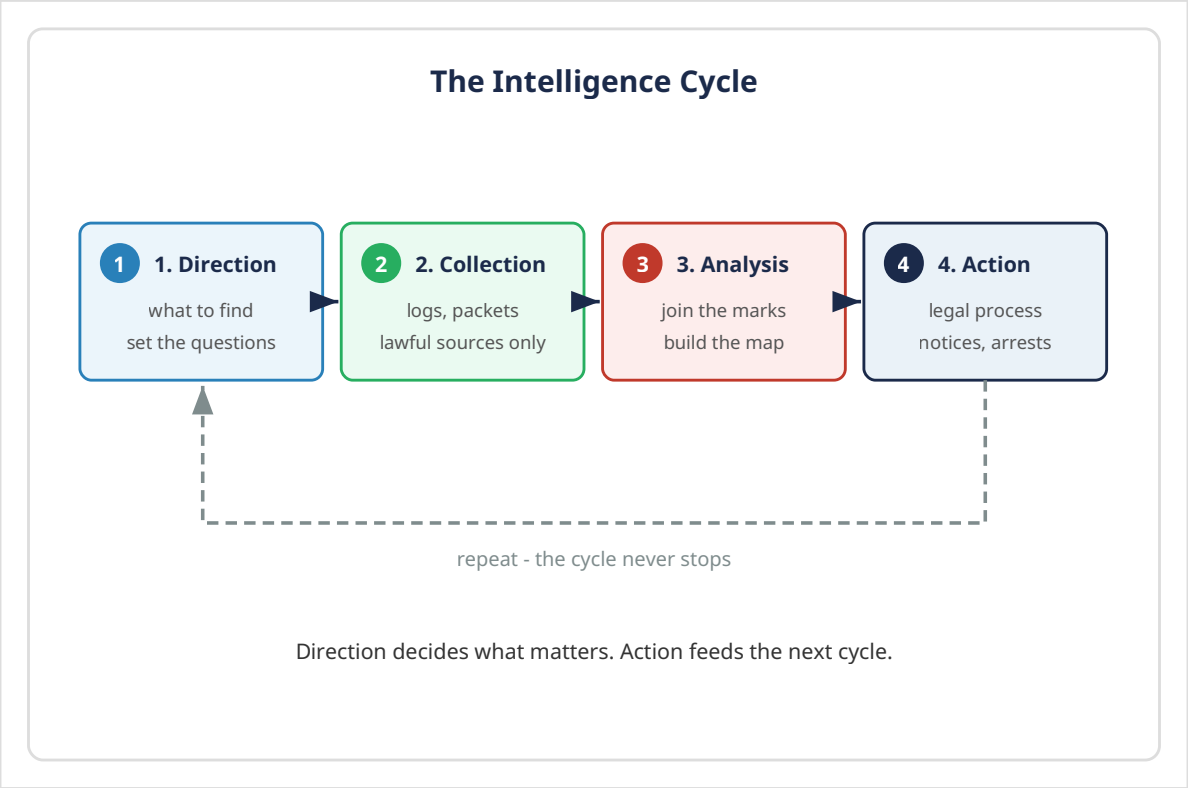
- Like money leaves a trail between bank accounts, **every connection leaves logs** at ISPs and server companies
- The criminal hides behind **VPNs, proxy servers and foreign servers** - AIIMS investigators learned this the hard way
- **Never rely on a single mark.** Join many marks together - only the full picture names the criminal
- > **Legal process is essential:** logs are demanded with a court order or BNSS notice - never poke through network data without permission.

Three Network Marks, One Map



- **IP + timestamp:** from which address, at what time
- **DNS records:** which servers the criminal looked up
- **Traffic patterns:** how much data, how long, which country

The Intelligence Cycle



- **Direction** decides the questions before a single log is pulled
- Collect only from **lawful sources** - everything else is inadmissible
- Analysis **joins the marks** into a map; action feeds the next cycle

Where Network Logs Come From - Part 1

You cannot join marks you never collected. **Four sources** hold the network's records - know them all.

- **ISP records (IPDR):** your internet provider knows which IP stood in whose name at what time
- **Server logs:** every website and email server records who connected, when, and what was asked - the access log
- Once you have the criminal's IP, **the ISP is asked whose name it stood in** at that exact time
- Logs do not last forever - companies delete them. **Act fast** or the mark is gone

Where Network Logs Come From - Part 2

Two more sources, and the one rule that keeps your evidence alive in court:

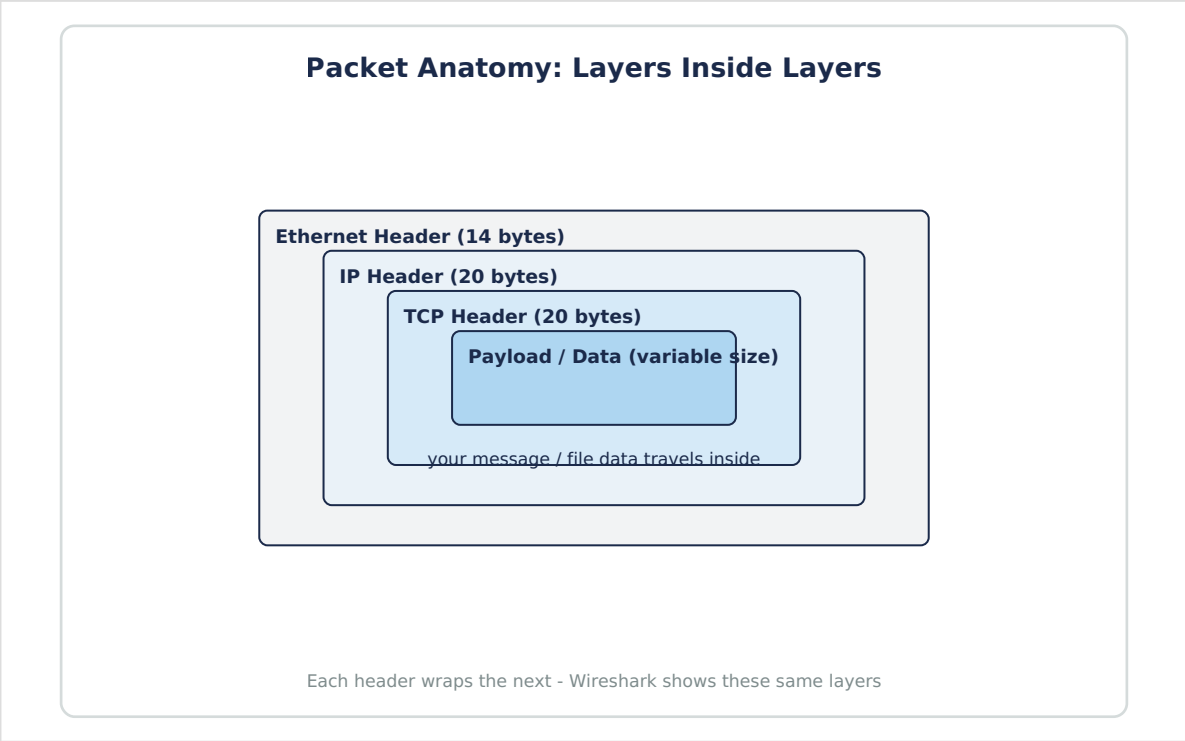
- **Email headers:** every email hides its journey - the Received lines show every server it passed through
- **Device logs:** routers, firewalls, office computers - an office firewall shows who inside connected where
- **BNSS s.94** (earlier CrPC s.91) lets the investigating officer demand records with a written notice
- A log demanded **without legal process does not hold up in court** - paperwork first, always

Packet, Proxy, Interception: Three Basic Words

Three words you will meet in every intelligence file. **Understand them in simple language**:

- **Packet:** data travels in small pieces - like a book divided into pages. Each packet carries sender and receiver addresses
- **Proxy:** a middleman server. The website sees the proxy's IP, not the criminal's - a wall, but with legal doors
- **Interception:** catching data in transit - like pulling a letter from the postman's bag. **Very sensitive in law**
- Criminals use proxies and VPNs to hide; **your counter is legal process**, not hacking back

Inside a Packet



- Wireshark captures **your network's packets** and shows them line by line - a free tool
- Encrypted traffic shows only the **route** - which server was talked to - never the inner conversation
- Safety: capture only on **your own network**, or with **written permission**

Proxy vs VPN vs Tor: Who Sees What

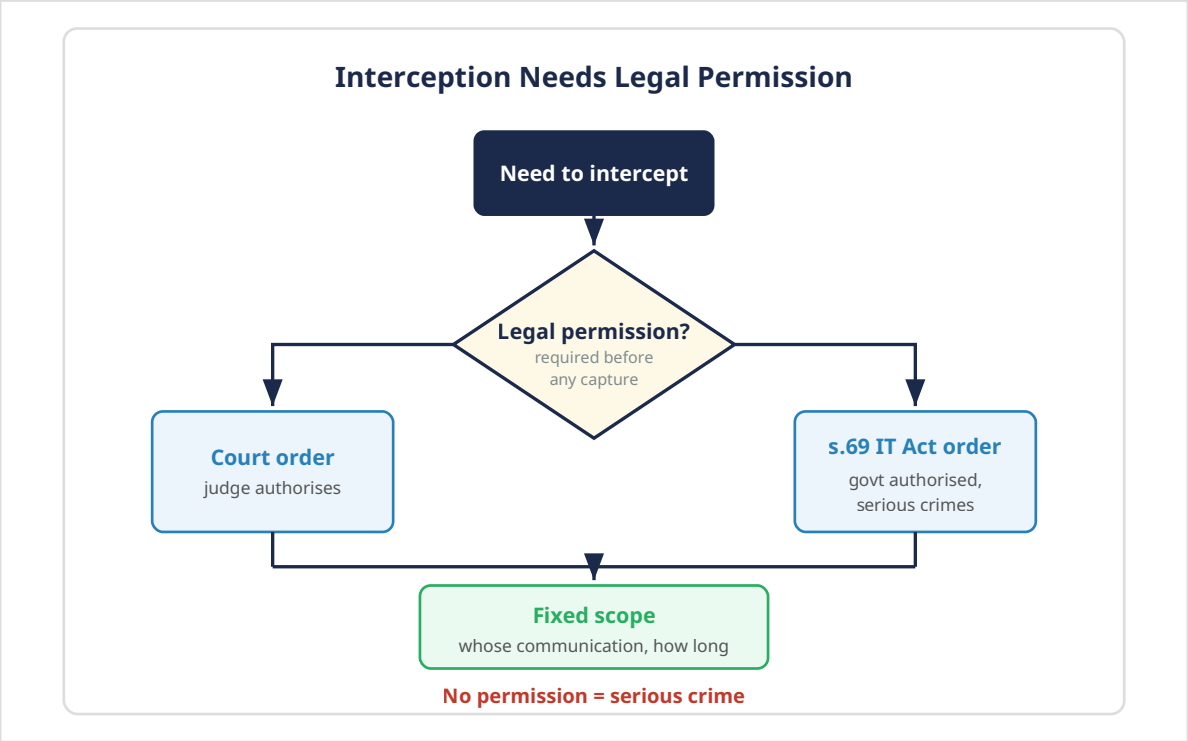
Proxy vs VPN vs Tor: Who Sees What

Question	Proxy	VPN	Tor
Website sees	proxy IP	VPN IP	exit node IP
ISP sees	proxy connection	VPN connection	Tor use
Weak point	proxy company logs	VPN company logs	exit node

No technique hides everything - every wall has a legal door.

- In all three, the website sees **a middleman's IP** - never the criminal's real one
- But in every technique **someone can still see something** - that someone is your legal door
- Using Tor is **not illegal** - criminal use is investigated through other evidence

Interception Needs Legal Permission



- Permission comes as a **court order**, or an **s.69 IT Act order** - government authorised, for serious crimes only
- Even with permission, the scope stays **fixed: whose, how long** - never cross it
- **Without permission**, interception is a serious crime - no exceptions

CVE: The Common Language of Weaknesses

CVE means **Common Vulnerabilities and Exposures** - the world's public list of known software weaknesses.

- Every known weakness gets a number, like **CVE-2021-44228** (Log4Shell) - one number, one meaning, worldwide
- Attackers exploit **well-known** weaknesses first - AIIMS had basic gaps like missing firewall policies
- **CVSS score 10.0 = Critical** - the most serious category, needing no password and no victim click
- When a server is hacked, check **which CVE the attacker used** - it tells you how he got in

APT: The Patient Enemy

APT means **Advanced Persistent Threat** - not ordinary hackers, but skilled groups fixed on one target for months or years.

- Goal is **espionage, not money** - slow, quiet, long-running attacks
- SideCopy APT (active since 2019): **spear-phishing emails** with fake PDFs that install spyware on opening
- They use **self-deleting routines** - yet marks survive in network logs, email headers and file fragments
- Investigator's rule: **watch patterns of months**, not one email or one file

VAPT: Find Your Weaknesses First

VAPT means **Vulnerability Assessment and Penetration Testing** - finding your own weaknesses before the criminal does.

- **Assessment** tells where the weaknesses are; **penetration testing** actually tries them to see if intrusion is possible
- Always done with the **owner's written permission** - testing anyone's system without permission is hacking
- Defensive scanning shows **open ports, old services with known CVEs**, and whether the firewall works
- Good intention does not make unauthorised testing legal - **permission is the line**

OWASP: The Web Attacker's Playbook

OWASP is an **open community on web application security**. Its famous OWASP Top 10 lists the ten biggest threats to web apps.

- Injection attacks, **broken access control**, incorrect security configuration - the usual suspects
- When a website is hacked, match the attack against this list to find **which well-known route** the attacker took
- Most website hacks use **known routes**, not new genius - the list shortens your investigation

Scanning Tools: Knock, Do Not Break

Scanning tools like **Nmap** collect information - which ports are open, which services run, which OS is installed. Like knocking on doors.

- Scan **only your own system**, or a system with the owner's **written permission**
- Scanning someone else's server - even 'just for information' - can be **unauthorised access**
- Use scanning **defensively**: your own systems, your client's systems - stop crime before it happens
- > **Investigators never scan the criminal's systems.** Your tools defend; your legal process attacks.

DNS and WHOIS: The Internet's Address Book

Two **free, fully legal** sources every investigator uses - reading them is not entering anyone's system.

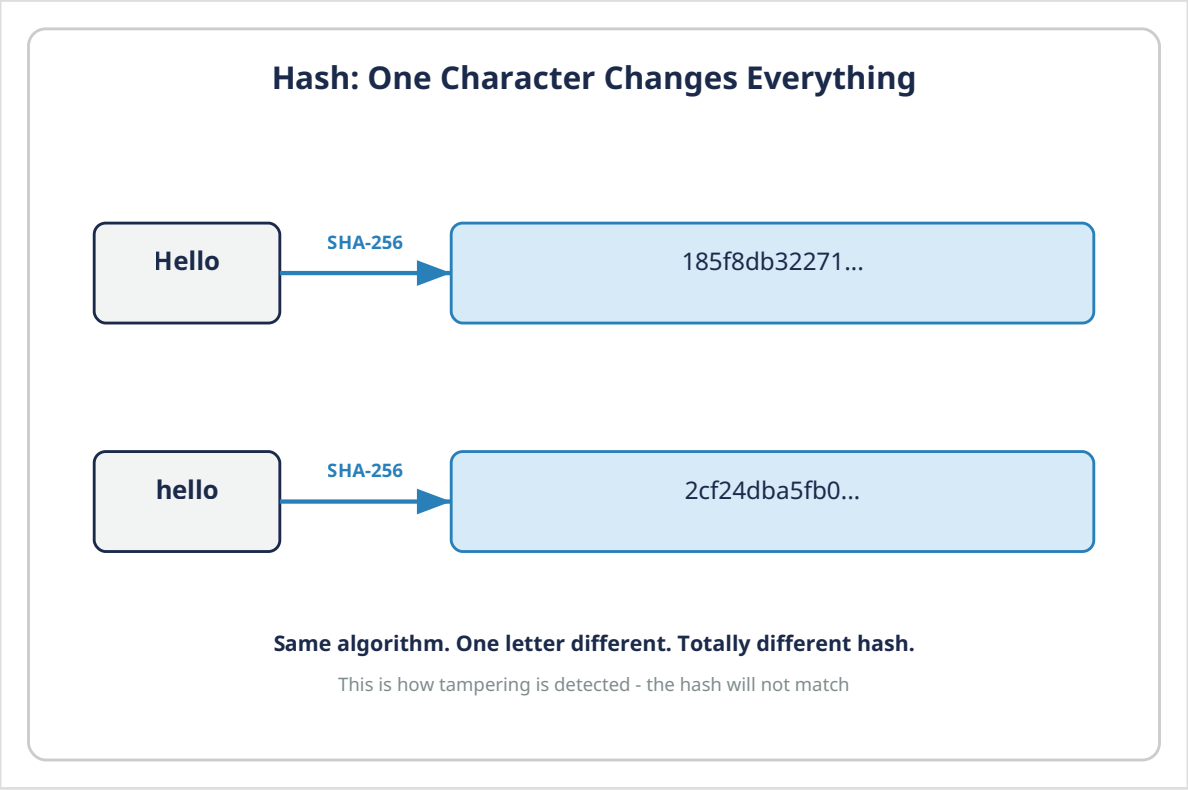
- **DNS** is the internet's phone diary - **nslookup** turns a name like example.com into an IP address
- In Wireshark you **see DNS queries flowing** - which domains the criminal looked up
- **WHOIS** gives domain ownership: creation date, expiry, registrar - a newly created domain in phishing is itself a mark
- Privacy rules may hide the owner's name, but **creation date and registrar stay available**

Cryptography, Hash, Digital Signature

Three words you will use **daily** in evidence work. Understand them in simple language:

- **Cryptography:** making information secret with a key - your WhatsApp and banking are protected by it; ransomware misuses it
- **Hash:** a file's **digital fingerprint** - change one character and the hash changes completely
- **Digital signature:** proves who really sent a document and that it was **not changed in transit**
- The hash is noted at collection and matched in court - **same hash means no tampering**

Catching Tampering with a Hash



- Change **one character** in a file - the SHA-256 hash changes completely
- Seize, **hash, record** - match again months later in court
- Different hash = **tampered**. Same hash = proof of purity

Anti-Forensics: The Criminal's Tricks

Anti-forensics means the tricks by which the criminal tries to erase his marks. Almost none of them work completely.

- Tricks: **deleting files, cleaning logs, timestomping** (fake timestamps), wiping, formatting the whole system
- But Windows keeps several diaries: **$MFT, USN journal, Prefetch** - clean one place, marks survive in another
- Deleted files leave **fragments**; cleaned logs survive on **backup servers**
- Your job: **find the leftover places** the criminal forgot

Encrypted Evidence: The One Rule

When encrypted evidence is found, the biggest rule is this: **never try to break it yourself**.

- Wrong tampering can **destroy the evidence itself** - and breaking encryption is a dead end for you
- Correct method: **secure it, hash it, maintain chain of custody**, hand it to a forensic expert
- The decryption key is demanded through **legal process** - from the accused or the service provider under court order
- > **Never break encryption yourself** - hash, secure, hand to expert.

First Response: Isolate and Secure

The AIIMS case gave one more lesson: the **NIC team immediately isolated** the infected servers so the infection would not spread.

- First response: **isolate the evidence** - disconnect from the network, then secure it
- Touching or rebooting a compromised system in a hurry can **wipe out evidence**
- Encryption does not hide everything - **who encrypted, when, connected to which key** are still questions to answer
- Examine later, in a controlled way - **haste is the evidence-killer**

Real Case: AIIMS Delhi Ransomware

23 November 2022 - **India's top hospital** frozen by cybercrime: appointment, billing, lab reports, all shut.

- Ransomware **encrypted 5+ servers**; doctors worked on paper for weeks - thousands of patients affected
- Attacker IP traced to **Hong Kong** - but VPNs mean an IP is a clue, not a destination
- CERT-In found **missing firewall policies**; Delhi Police IFSO registered an FIR under cyber-terrorism sections
- Lessons: **find the real route behind the VPN**, study CVEs, isolate evidence first
- Source Hindustan Times, The Indian Express

Real Case: The SideCopy APT Group

Reported 2026: an APT group active since **2019**, targeting India's educational institutions with spear-phishing.

- Earlier targets: **defence and government officials** - now universities
- Emails carry lures like **fake government circulars**; opening the fake PDF installs spyware
- The group uses **self-deleting routines** - but network logs and email headers survive
- Lessons: **watch patterns of months**; the email header is the first evidence
- Source The Hacker News

Common Mistakes Investigators Make

Most intelligence failures are **procedural**, not technical. Avoid these:

- Treating the **first IP found** as the criminal's real location - ignoring VPNs and proxies
- Collecting logs **without legal process** - evidence that dies in court
- Scanning or probing **the suspect's systems** yourself - unauthorised access, and it contaminates the case
- Touching a compromised system **before isolating it** - the infection spreads and evidence dies

Your Turn: Labs in This Module

Theory without practice is trivia. Chapter 21 has **four hands-on labs** - all on your own systems:

- **Lab 1:** Capture your own network traffic with Wireshark - find DNS queries and destination IPs
- **Lab 2:** Hash a file with SHA-256, change one character, and catch the tampering yourself
- **Lab 3:** Read CVE-2021-44228 (Log4Shell) on nvd.nist.gov - CVSS 10.0, Critical
- **Lab 4:** Fill the proxy-VPN-Tor comparison table - learn who can still see what
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> **Safety:** own network + written permission only. Reading CVEs is legal; building or trying exploits is not.

Key Takeaways

If you remember nothing else from Module 21, remember this:

- The criminal comes through the network - **learn its language**: packets, logs, DNS
- Three marks joined make a map: **IP + timestamp, DNS records, traffic patterns**
- **Never rely on one mark** - VPNs and proxies hide single clues
- CVE, APT, VAPT, OWASP - **speak the intelligence vocabulary**
- **Legal process first** - logs, interception, VPN records all need lawful authority
- Encrypted evidence: **never break it yourself** - hash, secure, hand to expert

Review Questions

- 1. Name the three network marks an investigator joins into the criminal's habit map - and say what each one tells you.
- 2. A trace leads to a VPN server's IP. The suspect claims this proves he was never involved. How do you answer, step by step?
- 3. What was the CVSS score of CVE-2021-44228, and why should an investigator check CVEs when a server is hacked?
- 4. Your unit wants to intercept a suspect's calls tonight, without any written order, 'to move fast'. What do you say and why?
- 5. You find an encrypted drive at a crime scene. List the four correct steps - and explain why breaking it yourself is wrong.

NEXT: MODULE 22

Digital Forensics Part 1 Evidence Speaks, If Handled Right

Five steps, chain of custody, write blockers - where cases are won or lost

Cyber Crime Investigator - Training Manual